Centers for Medicare & Medicaid Services (CMS)

RFI Response

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E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD–1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

ResolveSoft, a Women-Owned Small Business (WOSB), recommends that CMS stimulate developer interest by offering grant programs, technical challenges, and procurement pathways that prioritize inclusive small businesses. Specifically, we encourage CMS to expand the use of WOSB set-aside contract vehicles and sole source opportunities, which significantly reduce the administrative burden for small firms and accelerate time to value. Additionally, we advocate streamlined RFP processes that prioritize oral presentations, solutioning sessions, and past performance discussions over document-heavy submissions. These approaches foster meaningful engagement and allow WOSBs to demonstrate technical value and mission alignment early in the procurement lifecycle.

ResolveSoft is an ISO 9001:2015, 20000-1, and 27001 certified company, specializing in transforming digital offerings and delivering operations and maintenance support. We understand the critical missions of agencies such as the Department of Health and Human Services (HHS), including the Administration for Strategic Preparedness and Response (ASPR), Medical Reserve Corps (MRC), National Institutes of Health (NIH), Administration for Children and Families (ACF), Centers for Disease Control and Prevention (CDC), and the Assistant Secretary for Financial Resources (ASFR). ResolveSoft delivers full life-cycle Agile, Salesforce, and DevOps/DevSecOps transformations, with scalable capabilities in automation, data collection, and web application development.

Short-term actions CMS could take include expanding access to testing environments, simplifying vendor onboarding, and reducing barriers to access APIs and data sets. In addition, CMS can encourage WOSB participation and rapid solution development by offering incentives for building digital health tools, expanding access to real-time and de-identified datasets, and streamlining governance and compliance reviews. Reducing red tape and clarifying decision-making pathways to move digital products into production environments would directly benefit smaller, agile vendors like ResolveSoft.

Long-term strategies should include the establishment of dedicated innovation accelerators that provide mentorship, infrastructure, and access to CMS datasets for WOSBs and other small businesses. CMS could also support pre-qualified vendor pools for rapid tasking and foster collaborative development programs with built-in funding for pilot deployments and user research. Additional investment in open-source tooling, interoperable APIs, and reusable digital components will create a more equitable development landscape and lower technical barriers for participation in future CMS initiatives. Further, CMS should consider long-term programmatic investments that sustain innovation pipeline such as challenge grants, milestone-based funding, and technical assistance centers dedicated to supporting small and disadvantaged businesses.

TD–2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

CMS should expand available datasets to include supplemental benefits data, social risk indicators, and longitudinal beneficiary records.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

ResolveSoft believes that to supplement the Blue Button 2.0 API and increase its value for digital health applications, several additional datasets would be highly beneficial. These datasets would fill key gaps in clinical context, social determinants of health, and real-time care coordination:

1. Medicare Advantage (Part C) Claims Data

Why it matters: Blue Button 2.0 currently excludes Medicare Advantage claims, which cover ~50% of Medicare beneficiaries. Including this would provide a complete view of utilization, particularly for managed care.

Use case: Applications could compare traditional Medicare with Advantage plans, optimize benefit selections, and support whole-patient analytics.

2. Clinical EHR Data (C-CDA, FHIR Clinical Resources)

Why it matters: Claims data shows what was billed, not what actually happened in care. EHR data provides richer clinical details such as vitals, lab results, and clinical notes.

Use case: Improves chronic disease management, risk stratification, and transitions of care.

3. Social Determinants of Health (SDOH)

Why it matters: Health outcomes are deeply influenced by non-clinical factors like housing, transportation, food access, and social isolation.

Use case: Enhances predictive models, identifies at-risk populations, and allows for tailored interventions and referrals to community-based services.

4. Patient-Reported Outcomes (PROs)

Why it matters: Claims and EHRs miss the patient's voice, how they feel, function, and recover.

Use case: Enables holistic evaluation of care quality and effectiveness, especially in behavioral health and post-acute care.

5. Real-Time Admission, Discharge, Transfer (ADT) Feeds

Why it matters: Blue Button data has a lag. Real-time ADT alerts improve care coordination, reduce remissions, and support post-discharge follow-up.

Use case: Enables providers or care managers to intervene immediately after ER visits or inpatient stays.

6. Behavioral Health Data

Why it matters: Mental health and substance use treatment data are often fragmented or excluded. Integration is critical for whole-person care.

Use case: Combines physical and behavioral health data to better support patients with comorbid conditions.

7. Lab and Imaging Results

Why it matters: These are key to diagnostics and care decisions but are not captured in claims.

Use case: Enables clinical decision support, chronic condition tracking, and quality measure reporting.

8. Pharmacy Benefit Manager (PBM) and Prior Authorization Data

Why it matters: Claims only show what was dispensed; prior authorization delays and formulary restrictions are not visible.

Use case: Enhances medication adherence tracking and improves patient access planning.

Adding these data sets would transform Blue Button 2.0 from a retrospective billing snapshot into a comprehensive digital health record—unlocking better decision-making, predictive analytics, and value-based care optimization.

c. What obstacles prevent accessing these data sources today?

Current obstacles include inconsistent standards, authentication complexity, and limited cross-agency coordination.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

CMS and ASTP/ONC should consider APIs from SSA, NIH, and VA systems to promote cross-sector care coordination.

2. Digital Identity

TD–3. Regarding digital identity implementation:

a. What are the challenges and benefits?

ResolveSoft believes that challenges include technological access for underserved populations, difficulty using digital identity tools, and hesitancy around sharing personal information online. Many individuals—particularly older adults, rural residents, and low-income populations—face barriers such as limited internet access, lack of smartphones, or unfamiliarity with multi-step digital verification processes (e.g., facial recognition, document uploads). Even when access is available, users often struggle to complete identity verification successfully, leading to frustration and abandonment.

Additionally, there is widespread hesitancy to provide sensitive personal information over the internet, especially to unfamiliar third-party identity providers. Concerns about data misuse, identity theft, and lack of transparency in how information is stored or shared are common among Medicare beneficiaries.

A further challenge is the redundant requirement to verify identity across multiple health systems, applications, and government portals. This fragmented approach increases burden and reduces usability.

One potential solution is to implement a federated digital identity model, where a beneficiary completes identity verification once through a trusted platform (e.g., Login.gov or a NIST 800-63-3 compliant provider), and then use that verified identity to access multiple CMS-approved services and health systems. This “verify once, use everywhere” approach would reduce redundancy, streamline access, and improve user trust.

Benefits of digital identity include enhanced security, reduced fraud, and more personalized digital services. When implemented effectively, a secure digital identity reduces the risk of unauthorized access and identity-based fraud (e.g., fraudulent claims, impersonation). It also

allows for more seamless, personalized interactions with health systems, including real-time data access, telehealth, and digital consent management, all of which improve beneficiary experience and care coordination.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800–63–3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Requiring identity credentials would improve access control and auditability, ensuring that only authorized individuals access sensitive health data and enabling traceable, secure interactions across CMS systems. This approach helps reduce fraud, supports compliance with federal privacy standards, and facilitates precise data sharing with verified individuals.

However, it could also result in delayed access for users who encounter issues during the identity verification process. Many beneficiaries, particularly those less familiar with digital tools may face challenges uploading documents, completing biometric steps, or navigating multi-factor authentication processes. These obstacles could deter timely access to care-related systems, especially when real-time verification is required. To mitigate this, CMS could offer streamlined fallback options, such as assisted identity verification or one-time verification with persistent credentials that can be reused across systems.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

OpenID Connect adoption would standardize protocols across vendors and simplify multi-platform integration.

3. Technical Standards and Certification

TD–4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

ResolveSoft encourages CMS to offer higher reimbursement rates or bonus scoring for solutions that adopt FHIR-based, open APIs. Additionally, developer outreach programs and open-source reference implementations would be helpful.

TD–5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A publicly maintained FHIR endpoint directory would significantly promote interoperability by enabling secure, reliable discovery of healthcare data exchange points across payers, providers, and networks. Such a directory would act as a national index of where and how to connect to electronic health records (EHRs), payer APIs, and data-sharing networks—critical infrastructure for enabling scalable, real-time data exchange.

CMS is uniquely positioned to lead this effort due to its broad reach across the healthcare ecosystem and its regulatory authority over Medicare and Medicaid providers and payers. By owning and managing the directory, CMS can ensure consistent technical standards, equitable participation, and uniform compliance with national interoperability goals. This leadership would also prevent fragmentation and duplication of effort that could arise if multiple private or regional entities attempted to maintain disconnected directories.

Furthermore, CMS should provide free access to this directory, especially for early-stage innovators, startups, and Women-Owned Small Businesses (WOSBs). Commercially operated directories often impose fees or licensing restrictions that disproportionately impact small businesses—many of whom are driving innovation in digital health. Free, public access would

level the playing field, encouraging broader participation in federal health IT initiatives and enabling smaller vendors to develop interoperable tools that connect seamlessly across the national healthcare infrastructure.

In summary, CMS-led, publicly accessible FHIR endpoint infrastructure is critical to advancing nationwide data exchange, reducing onboarding friction, and enabling inclusive innovation across the healthcare landscape.

TD–6. What unique interoperability functions does TEFCA perform?

a. What existing alternatives should be considered?

TEFCA promotes a unified national exchange but must streamline redundant functions across DirectTrust, CommonWell, and Carequality.

b. Are there redundant standards, protocols or channels or both that should be consolidated?

CMS should work with ONC to align existing frameworks and reduce technical fragmentation.

TD–7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

a. Does it contain the full extent of data elements you need?

USCDI has improved interoperability by standardizing core data elements across healthcare systems, enabling more consistent and accessible exchange of patient information. It has facilitated better coordination of care, improved data quality, and enhanced support for clinical decision-making, especially in primary care and acute care settings.

However, gaps remain particularly in behavioral health and social care data—which limits the ability of providers and digital tools to deliver whole-person care. For example, USCDI currently includes limited structured fields for behavioral health conditions, treatment plans, or therapy notes. Critical data such as psychiatric evaluations, substance use history, and mental health crisis events are often stored in unstructured formats or siloed in non-integrated systems, making it difficult to exchange or act on that information consistently.

In social care, while USCDI Version 3 includes some Social Determinants of Health (SDOH) elements, such as housing and food insecurity, it lacks comprehensive coverage of care coordination activities, social service referrals, program enrollment status, and outcomes of community-based interventions. These data points are essential for addressing non-clinical factors that drive health disparities and influence outcomes—particularly for Medicare and Medicaid populations.

The lack of standardized, interoperable data in these domains creates blind spots for care teams, researchers, and policymakers. Without this information, risk assessments may be incomplete, interventions may be mistimed or misaligned, and value-based care models may fall short of addressing patient needs.

To close these gaps, CMS and ONC should work with behavioral health providers, social service organizations, and advocacy groups to expand USCDI to include more structured behavioral health data (e.g., DSM-5 conditions, care plans, medication adherence) and social care elements (e.g., resource utilization, referral resolution, caregiver support). This expansion would enable more equitable, coordinated, and outcomes-driven healthcare delivery across settings.

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

See response to question A above.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

See response to question A above.

d. Given improvements in language models, would you prefer a non- proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

Language model-based inputs could significantly increase data coverage and completeness particularly for unstructured or under-documented domains such as behavioral health, social care, and patient-reported outcomes. These AI-driven tools can extract meaningful data from free-text clinical notes, patient narratives, and other non-standardized inputs, helping to populate structured fields that are missing or inconsistently captured in EHRs and claims systems.

For example, natural language processing (NLP) and large language models (LLMs) can identify medication adherence issues, functional status, psychosocial risk factors, or care coordination notes embedded in provider documentation. This capability would be especially valuable for supplementing gaps in USCDI-aligned datasets without requiring extensive workflow changes or manual abstraction by clinicians.

However, the use of language models in healthcare data pipelines must be accompanied by strong governance and guardrails to ensure safety, accuracy, privacy, and fairness. Key considerations include:

Data provenance: Ensuring outputs from language models are traceable, transparent, and clearly marked as AI-generated or AI-extracted.

Validation and accuracy: Establishing model performance benchmarks for healthcare-specific domains and use cases, with clinician review or approval loops where appropriate.

Bias mitigation: Monitoring and adjusting for disparities in how language models interpret data from underrepresented populations or non-standard language patterns.

Privacy and compliance: Enforcing HIPAA compliance and preventing unintended exposure of protected health information (PHI) during training, inference, or storage.

When implemented responsibly, language models can serve as powerful tools for enhancing interoperability by transforming previously inaccessible or underutilized text into actionable, structured data elements. This would enrich datasets used for care coordination, quality reporting, risk adjustment, and research—especially in areas where standardized coding has historically lagged.

CMS and ONC could further support this innovation by developing guidance for AI-augmented data capture, encouraging pilot programs, and including AI-extracted fields as extensions to USCDI under defined use cases and validation thresholds.

TD–8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

ResolveSoft recommends emphasizing testable API conformance, integration with clinical decision support (CDS) tools, and ease-of-use in real-world provider workflows as key pillars of an effective certification framework under the ONC Health IT Certification Program.

First, testable API conformance ensures that health IT systems not only claim to support interoperability standards (such as FHIR or SMART on FHIR), but actually implement them correctly, consistently, and securely. Too often, "certified" APIs vary in implementation, leading to costly custom integrations and interoperability gaps. CMS and ONC should enforce rigorous, automated testing protocols—preferably through publicly available tools—to verify that APIs can exchange structured data reliably across vendors and use cases. Certification should include real-world use tests, not just laboratory validation.

Second, certification should promote seamless integration with clinical decision support (CDS) tools. These tools are essential for translating raw data into actionable insights at the point of care—helping clinicians make better decisions and improving outcomes. Certified systems should demonstrate the ability to ingest standardized data from external sources and trigger CDS alerts or care recommendations in line with clinical guidelines. APIs should support bi-directional data exchange so that CDS platforms can both receive and contribute to patient records.

Finally, ease-of-use in live provider workflows is critical. Certified systems must go beyond technical compliance to support real-world usability. This includes minimizing disruption to clinical tasks, ensuring intuitive user interfaces, reducing alert fatigue, and enabling patient-centered views of data. Certification criteria should incorporate human-centered design principles and usability testing, especially for modules used in high-volume care environments such as primary care, emergency rooms, and long-term care facilities.

Together, these priorities ensure that certification is not just a checkbox exercise but a meaningful signal of functional, secure, and user-friendly technology that supports interoperability, improves care delivery, and reduces administrative burden.

TD–9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

An API-first certification model would promote real-time interoperability by shifting the focus of health IT validation from static software functions to dynamic, standards-based data exchange capabilities. Rather than simply certifying whether a system includes a certain feature (e.g., a medication list or clinical note viewer), API-first certification ensures that data behind those features can be accessed, shared, and used across systems through modular, well-defined, and consistently implemented interfaces.

This approach aligns with how modern digital ecosystems operate—relying on application programming interfaces (APIs) to power real-time connectivity, automation, and user-centered design. By making APIs the foundation of certification, CMS and ONC would promote interoperability not just in theory, but in practical, measurable ways. Systems that pass certification would need to demonstrate the ability to:

Enable external access to key clinical and administrative data elements (e.g., problems, medications, encounters, labs, SDOH data, etc.) using FHIR or equivalent open standards.

Support bi-directional data exchange, allowing outside tools—such as care coordination platforms, third-party apps, and analytics engines—to both read and write data securely.

Work reliably in real-world environments, not just test beds, with validated conformance to implementation guides and versioning standards.

For providers and patients, this results in faster, more seamless access to health information across settings and applications, reducing duplication of effort and improving continuity of care. For developers and small businesses like ResolveSoft, it lowers the integration burden and encourages innovation by ensuring a consistent technical foundation across certified systems.

An API-first approach would also better support CMS policy goals, including price transparency, patient access, and value-based care, by enabling the real-time flow of actionable information into tools that drive smarter decisions at the point of care.

In sum, API-first certification future-proofs the health IT ecosystem, promotes vendor accountability, and aligns federal certification programs with modern, scalable digital infrastructure principles.

b. What would be the drawbacks?

While **API-first certification promotes transparency and scalable interoperability**, it requires **robust guardrails, implementation consistency, and support mechanisms** to ensure that the model doesn't unintentionally exclude smaller vendors, introduce security vulnerabilities, or burden providers. Careful policy design, including **phase-in periods, technical assistance, and cost-sharing options**, can help mitigate these challenges.

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

Certification should require read/write API access across both structured and unstructured chart data to fully enable interoperability and support real-world clinical and administrative use cases. While structured data—such as medications, labs, diagnoses, and procedures—forms the backbone of interoperability today, much of the most valuable clinical information still resides in unstructured formats, including physician notes, care plans, discharge summaries, scanned documents, and correspondence.

Limiting certification to read-only access or structured fields restricts the ability of external applications, care teams, and health data networks to participate in the full life cycle of health information exchange.

For example: Read-only access prevents tools from writing back updated medication lists, allergy data, or patient-generated observations—hindering collaborative care models.

Exclusion of unstructured data means decision support tools, research platforms, and AI-driven analytics solutions cannot access a large portion of clinically relevant insights, such as narrative notes documenting social risk factors, mental health assessments, or end-of-life preferences.

Requiring read/write capabilities across both data types would ensure that certified systems can act as true interoperability platforms—not just data silos with limited visibility or editing functionality. This would also enable:

Enhanced care coordination: Tools could update patient charts with care team communications or community referrals.

Closed-loop workflows: External apps could contribute structured inputs back into the record (e.g., screening results, eConsent, patient-reported outcomes).

Richer analytics: NLP tools and language models could extract insights from free-text and feed structured summaries into EHRs or dashboards.

To support this, certification should define:

- Required API endpoints for both read and write access,
- Minimum data elements and document types to expose (e.g., C-CDA, clinical notes, PDFs),
- Safeguards for access control and audit logging, ensuring write permissions are secure and appropriate,
- Support for extensions like SMART on FHIR write-back scopes and granular user authorization frameworks.

Ultimately, enabling full read/write interoperability is essential to transforming EHRs from passive repositories into active, collaborative platforms that empower patients, providers, payers, and developers alike.

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

CMS should provide funding for early adopters and penalize non-participation.

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

Bulk data-enabled EHRs can streamline reporting especially for quality, performance, and population health if paired with CMS-hosted dashboards and simplified, standardized measure definitions. Under the current system, providers and health systems often face fragmented reporting requirements, inconsistent measure interpretations, and significant manual data abstraction and submission burdens. These inefficiencies drive up administrative costs and reduce time spent on patient care.

By enabling bulk FHIR data exports (via the Flat FHIR / Bulk Data Access API standard), EHRs can support automated, scalable extraction of large datasets—such as all patients' lab results, clinical encounters, or chronic condition status—on a scheduled or ad-hoc basis. When this capability is integrated with centralized CMS dashboards, providers could:

- Visualize performance in real-time against required measures,
- Benchmark outcomes across peer groups or regions,
- Receive proactive feedback on gaps in care or reporting,
- Identify patients who meet or fail quality thresholds (e.g., for HEDIS, MIPS, or ACO measures).

To fully realize this vision, several capabilities are required:

Standards-Based Bulk Data APIs - EHR systems must implement bulk export in compliance with ONC standards, supporting predefined resource types and patient cohorts.

Harmonized, Simplified Measure Definitions - CMS should consolidate overlapping measures and publish them using machine-readable logic (e.g., Clinical Quality Language - CQL) that EHRs and dashboards can interpret consistently.

CMS Dashboards with Secure Integration - CMS should provide cloud-based portals that ingest bulk FHIR files directly or link to API endpoints, enabling automated analysis and visualization.

Incremental and Delta Reporting Support - EHRs should support partial exports and change tracking, so only updated records are sent—reducing system strain and improving timeliness.

Robust Privacy and Access Controls - Role-based access and encryption are necessary to ensure compliance with HIPAA and prevent misuse of patient-level data.

Key Benefits of This Approach:

- Reduces Administrative Burden: Automates tedious manual chart reviews and data submissions, freeing up staff resources.
- Improves Accuracy and Timeliness: Minimizes transcription errors and enables up-to-date measure tracking.
- Enables Proactive Interventions: Real-time insights empower providers to take corrective actions before reporting deadlines.
- Fosters Equity and Transparency: Dashboards can highlight disparities in care and help target improvement initiatives.
- Encourages Participation in Value-Based Programs: Easier reporting makes it more feasible for small practices and WOSBs to participate in CMS incentive programs.

TD–10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj–11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

ResolveSoft supports requiring all EHR vendor APIs to allow seamless access and exchange without paywalls or custom interfaces. CMS should monitor and enforce this condition strictly.

TD–11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or ''EHI export'' for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export.

Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

CMS should standardize API formats for Electronic Health Information (EHI) export to promote consistency, usability, and interoperability across the healthcare ecosystem. Currently, the EHI export requirement under the ONC Health IT Certification Program allows health IT developers broad discretion in how they implement this capability. As a result, different vendors have adopted varying data formats, export packaging methods, metadata structures, and delivery mechanisms.

This lack of uniformity creates significant challenges for data recipients—such as patients, providers, researchers, payers, and third-party app developers—who must build custom solutions to interpret each vendor's output. Without standardized formats, exported data can be incomplete, poorly structured, or difficult to parse and integrate into downstream workflows.

By standardizing API formats for EHI export, CMS would ensure that data recipients can consistently:

- Access all required data elements in a predictable, machine-readable format;
- Incorporate exported data into existing FHIR-based platforms without complex transformation pipelines;
- Enable seamless patient-directed exchange, allowing individuals to port their health information across systems and applications effortlessly;
- Accelerate innovation, especially for small and emerging technology vendors who lack the resources to accommodate multiple proprietary formats.

Standardization would also benefit providers by simplifying compliance, onboarding, and integration with external applications or reporting platforms. This is particularly important for multi-facility systems and Accountable Care Organizations (ACOs) that work with various EHR vendors and need cohesive, scalable data aggregation strategies.

Additionally, consistency in export format is essential for regulatory enforcement and oversight. A standard baseline would allow CMS and ONC to verify that certified health IT systems are truly providing comprehensive and accessible EHI exports—not just fulfilling the requirement in name only.

Key features that CMS should standardize include:

- Use of FHIR bulk data (Flat FHIR) or structured C-CDA as export formats;
- Uniform metadata schemas and export file organization;
- Required data elements (e.g., clinical notes, labs, medications, imaging);
- Support for both single-patient and population-level exports;
- Guidance on delivery mechanisms

b. Are there specific workflow aspects that could be improved?

Export usability can improve with consistent metadata schemas and visual aids.

c. Should CMS consider policy changes to support this capability's use?

CMS should incentivize implementation via reporting bonus programs.

4. Data Exchange

TD–12. Should CMS endorse non- CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

ResolveSoft believes that CMS should endorse non-CMS data networks that meet transparent, measurable criteria for equity, privacy, and real-time access to promote a more trusted, interoperable, and inclusive health data ecosystem. As healthcare data networks—such as Health Information Exchanges (HIEs), Qualified Health Information Networks (QHINs), and commercial data aggregators—play an increasingly vital role in care coordination and analytics, CMS endorsement can serve as a powerful signal of quality, trust, and alignment with national health goals.

To make this endorsement meaningful and impactful, CMS should define a clear set of metrics and qualifications that data networks must meet, including:

Equity: Demonstrated efforts to ensure participation and service coverage in underserved, rural, and historically marginalized communities; support for multilingual data exchange and social determinants of health data sharing.

Privacy and Security: Adherence to HIPAA and 42 CFR Part 2 standards, robust access controls, audit capabilities, data minimization practices, and transparency in data use policies.

Real-Time Access: Support for modern APIs (e.g., FHIR, TEFCA-aligned interfaces), low-latency data exchange, and high system uptime to enable timely clinical decision-making and patient engagement.

Governance: Transparent governance models that include representation from diverse stakeholders including providers, patients, and small vendors.

Endorsing networks that meet these criteria would help standardize expectations across the ecosystem and reduce variability in performance and trustworthiness. Many organizations—especially small healthcare providers, startups, and WOSBs—lack the capacity to vet network partners for interoperability readiness or compliance. CMS endorsement would simplify partner selection and de-risk data sharing arrangements, accelerating innovation and adoption.

To operationalize this approach, CMS should establish a public trust registry—a centralized, regularly updated list of endorsed data networks that includes:

Certification status and qualifying metrics;

Supported standards and APIs;

Geographical and demographic coverage;

Access models and pricing structures;

Performance benchmarks (e.g., latency, data completeness, patient matching rates).

This registry would act as a transparent, government-backed resource for providers, payers, technology developers, and patients to identify and engage with networks that are secure, standards-compliant, and equitable.

Ultimately, CMS endorsement paired with visibility in a public registry—would create market incentives for data networks to invest in compliance, openness, and inclusivity, while giving stakeholders greater confidence in the reliability and fairness of the partners they choose to connect with.

TD–13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

a. What are the primary obstacles to this?

Barriers include lack of standardization and variable EHR vendor participation.

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

USCDI provides a well-defined, structured foundation for data interoperability, but full Electronic Health Information (EHI) allows for the clinical nuance and context that are often essential to delivering whole-person care. While USCDI helps standardize the most common and critical data elements—such as medications, allergies, immunizations, and problems it does not encompass the full breadth and complexity of a patient's health record.

For example, behavioral health assessments, social needs documentation, care coordination notes, provider-to-provider correspondence, and patient narratives are often stored in unstructured or non-standard formats that fall outside the USCDI scope. These components may lack standard vocabularies or consistent capture methods but can contain critical insights into a patient's care needs, preferences, and lived experience.

By contrast, full EHI, as defined by the 21st Century Cures Act, includes all electronic Protected Health Information (ePHI) in a Designated Record Set, regardless of whether it conforms to a standard structure. This broader view of the medical record supports more comprehensive use cases, including advanced analytics, quality measurement, behavioral health integration, and legal or compliance inquiries.

To balance these strengths, flexible APIs should be capable of supporting both structured USCDI data and unstructured or semi-structured full EHI content. This includes:

- Granular FHIR resources for USCDI-aligned data elements (e.g., Conditions, Observations, Medications);
- Support for unstructured documents via DocumentReference or Binary resources, including clinical notes, scanned files, and PDFs;
- Smart rendering tools that present structured and unstructured content together in context for providers and patients;
- Clear metadata tagging and provenance tracking to distinguish between validated structured data and less formal content.

Supporting both USCDI and full EHI through flexible APIs allows health systems and app developers to innovate responsibly leveraging standardization where it exists, while preserving access to rich, unstructured content when it's needed. It also ensures compliance with federal data access mandates while empowering more complete, human-centered care.

CMS and ONC should encourage API designs that allow clients to selectively query structured elements for automation and reporting, or retrieve full clinical content for deep review and analysis depending on the use case. This hybrid approach enables interoperability that is both technically robust and clinically meaningful.

TD–16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

a. Do current rules encourage scalable network participation?

Current rules provide a foundation for scalable participation, but do not go far enough to fully incentivize or operationalize it—particularly for small providers and innovative technology vendors. TEFCA and the 21st Century Cures Act promote the idea of nationwide, standardized health information exchange, but voluntary participation, variable implementation of FHIR APIs, and uneven enforcement of information blocking rules create inconsistency in the ecosystem.

Many providers, payers, and networks still rely on proprietary systems and localized agreements, and are not compelled to join shared networks unless participating in specific federal programs. The lack of mandates, financial incentives, and infrastructure subsidies makes it difficult for resource-constrained stakeholders—including community health centers, rural hospitals, and WOSBs—to integrate at scale.

Moreover, existing certification and compliance frameworks often focus on technical conformance rather than actual connectivity or usage metrics. This can result in certified systems that meet standards in theory but fail to support scalable, real-world interoperability in practice.

To encourage broader and more equitable participation in shared network infrastructure, CMS and ONC could:

- Tie payment incentives or program eligibility to active participation in TEFCA or other certified exchanges;

- Provide technical assistance and onboarding support to small and under-resourced organizations;
- Offer grants or shared infrastructure funding to reduce integration costs;

Require minimum interoperability capabilities (e.g., FHIR-based bulk data export, record location services, and patient identity resolution) as part of certification and provider compliance.

b. What changes would improve alignment (for example, API unification, reciprocal access)?

CMS should mandate reciprocal access standards and support shared infrastructure via public-private partnerships.

TD–17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

5. Compliance

TD–18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non- responsiveness as well as situations that may cause a failure or unusable response?

ResolveSoft has not experienced information blocking.

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

TD–19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

Despite recent regulatory progress, many hospital and payer price transparency files remain technically compliant but practically unusable. Pricing data is often published in large, flat files—such as CSVs or JSON documents—with inconsistent formatting, lack of standardized coding (e.g., HCPCS, CPT), or ambiguous service descriptions. These files are difficult to parse and interpret, especially for patients and third-party application developers without technical expertise.

Additionally, data is not updated frequently enough to reflect real-time pricing changes, coverage updates, or negotiated rate fluctuations. This stale or incomplete data undermines trust, creates confusion, and limits the ability of patients and providers to make informed decisions based on accurate cost projections. Users are left with a fragmented view of what a service will actually cost them, leading to surprise billing and diminished consumer engagement.

b. Which workflows would benefit most from functional price transparency?

Price transparency becomes especially valuable in workflows where patients and providers are actively planning care and making cost-related decisions. Two of the most critical use cases are:

Prior authorization: Real-time visibility into procedure pricing, covered services, and expected patient cost-sharing can help determine whether care is financially accessible—and allow for timely appeals or alternative care planning if needed.

Patient cost estimation tools: These tools, often used in scheduling, pre-registration, or provider-patient conversations, rely on timely access to pricing data to offer patients realistic out-of-pocket cost estimates. Without accurate and accessible pricing data, patients are forced to make decisions blindly, potentially delaying care or facing financial hardship.

Improving price transparency in these workflows enhances patient trust, provider efficiency, and payer-provider collaboration.

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

To modernize and scale price transparency, CMS should promote the development and adoption of standardized FHIR-based APIs for pricing data. These APIs would allow authorized systems to request real-time cost estimates for services, taking into account:

- Procedure codes or clinical service bundles;
- Payer-negotiated rates;
- Patient-specific benefits and deductible status;

Geographic location and provider-specific contracts.

FHIR-based pricing APIs enable automated, accurate cost calculations during provider workflows and patient self-service interactions. CMS support could include technical implementation guides, certification criteria, and testing environments, similar to how CMS has promoted the use of FHIR for claims and clinical data.

d. What would further motivate solution development?

To jumpstart creative solutions and lower barriers to entry, CMS should launch targeted funding opportunities—such as innovation challenges, Small Business Innovation Research (SBIR) grants, or public-private accelerators—to develop consumer-facing and provider-integrated pricing tools.

Prize competitions could reward solutions that demonstrate:

- Real-time cost estimation accuracy;
- Integration with scheduling or prior authorization tools;
- Patient-centered design and accessibility;
- Interoperability across payers and providers.

Such programs would incentivize startups, WOSBs, and academic partners to build scalable and equitable solutions for price transparency—supporting the goals of informed decision-making and healthcare affordability.